PHYS 6031
Spring 2024
Homework 2
Due: Friday 2 February 2024 at 4:00 pm in my Clippinger mailbox

1. Determine the capacitance of two concentric cylinders of length L and radii a and b, with $b > a$. You may assume that the cylinders are "long" – in other words that $L \gg a, b$.

2. A model for the electrostatic potential of a nucleus is that it can be considered a sphere of uniform volume charge density. Assuming a total charge of Q and that the radius of the sphere is R, determine the following.

(a) The electrostatic potential $\Phi(r)$, where $0 \leq r < \infty$, $r = 0$ at the center of the nucleus, and $\Phi(r \to \infty) \to 0$.

(b) The stored energy, using Eq. (1.53) of Jackson.

(c) The stored energy, using Eq. (1.54) of Jackson.

(d) What happens to the stored energy when $R \to 0$?

3. Some practice with an infinite line charge.

(a) What is the electric field on an infinite line charge that has a constant charger per unit length of λ?

(b) What is the electrostatic potential in this case? Is it possible to make the potential vanish at infinity?

(c) Consider an infinite conducting cylinder of radius a that is grounded. Suppose there is an infinite line charge with a constant charge per unity length λ that is located inside the cylinder, such that it is parallel to the cylinder, but displaced from the central axis by an amount b, with $b < a$. Determine the electrostatic potential everywhere inside the cylinder. I suggest using the method of images, by placing another line charge outside the cylinder.

4. Using the method of images, consider the problem of a point charge q **inside** a hollow, grounded, conducting sphere of inner radius a. Find:

(a) The potential inside the sphere. Your solution must include a proof that the potential vanishes on all points on the surface of the sphere.

(b) The induced surface-charge density on the inside surface of the sphere.